Chapter 4 | Blanchard, Macroeconomics 9e

Session 1: Money Demand + Interest Rate Determination

Session 2: Banks, Reserves & Liquidity Traps

73-103 Principles of Macroeconomics | CMU-Q

Carnegie Mellon University Qatar

Financial Markets I

Why Money & Interest Rates Matter

Why Should You Care About Financial Markets?

The real-world impact on your daily life

- **Interest rates affect everything: mortgages, car loans, business investment**
- **Financial markets channel savings into productive investment**
- **Central bank decisions ripple through the entire economy**
- **Financial markets: bonds, stocks, currencies, mutual funds**

Example: Global Bond Market
~$130 Trillion
Larger than all stock markets

KEY INSIGHT Financial markets are the transmission mechanism between central bank policy and the real economy.

SECTION 4-1

The Demand for Money

Why do people hold money instead of interest-bearing assets?

- The money vs. bonds tradeoff
- Determinants of money demand
- The money demand function

Money vs. Bonds: The Fundamental Tradeoff

The simplifying assumption: only two financial assets

MONEY		BONDS
✓ **Yes: spend it now (liquid)**	**Use it for transactions?**	✗ No: must sell it first
✗ No: 0% return	**Pays interest?**	✓ **Yes: earns the interest rate *i***
✓ **None: value fixed in dollars**	**Risk of price change?**	✗ Yes: its price can change
Cash, checking accounts *Qatar: QAR cash, QNB checking*	**Examples**	T-bills, government bonds *Qatar: sovereign bonds*

Example: $10,000 for one year, *i* = 5%

As money: $10,000, and you can spend it any time

As a bond: $10,500 next year (+ $500 interest)

Opportunity cost of holding money = *i* × $10,000 = $500

LIQUIDITY convenience ⟷ **RETURN** interest

Higher *i* → convenience costs more → people hold less money (next slide).

KEY INSIGHT The choice between money and bonds depends on the tradeoff between liquidity (convenience) and return (interest).

What Drives Money Demand?

Two key determinants

1. Level of Transactions (Nominal Income, $Y)

- More income = more spending = need more money
- Money demand increases proportionally with $Y
- Think: Holiday shopping season vs. January

↑ $Y → ↑ Md

2. Interest Rate on Bonds (i)

- Higher i = higher opportunity cost of holding money
- People shift wealth from money to bonds
- Money demand decreases as i rises

↑ i → ↓ Md

The Money Demand Function

Formalizing the relationship

$$Md = \$Y \times L(i)$$

Money Demand = Nominal Income × Liquidity Function

Md
Money Demand
(how much money people want to hold)

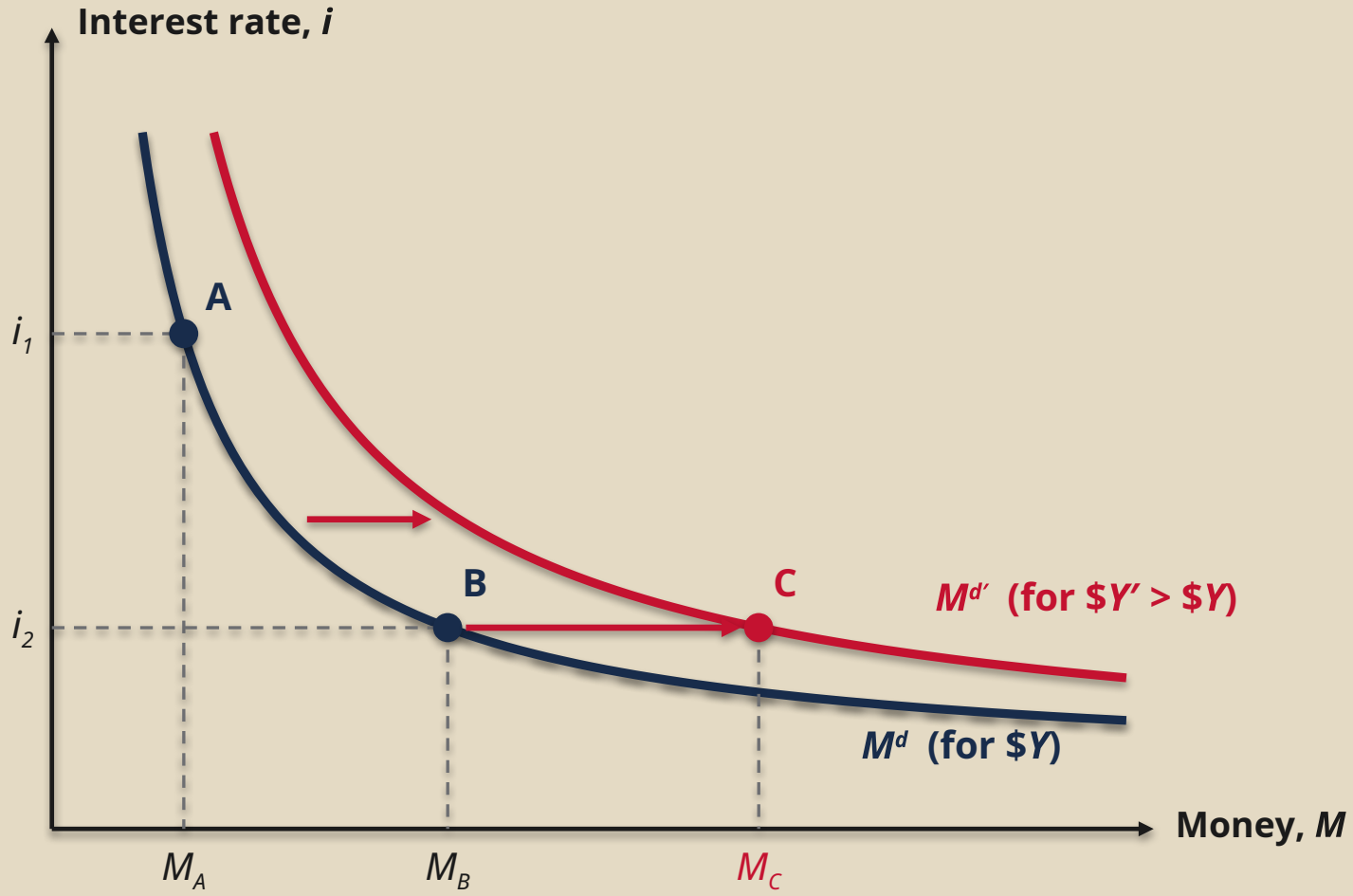
$Y
Nominal Income
(proportional: more income = more demand)

L(i)
Liquidity Function
(decreasing in i: higher i = less demand)

KEY INSIGHT L(i) is a decreasing function: as interest rates rise, people hold less money and more bonds.

Figure 4-1: The Demand for Money

Money demand as a function of the interest rate



Downward slope: A → B

Higher *i* = higher opportunity cost of holding money. As *i* falls, people hold more money.

Scenario: income rises ($Y ↑)

More income = more transactions = more money demanded at every interest rate.

Higher income shifts M^d right: at every interest rate, people want to hold more money.

Semantic Traps: Money ≠ Income ≠ Wealth

Getting the vocabulary right

Illustration: one household

Income
$5,000 per month

FLOW · measured per period

Saving
$1,000 per month

FLOW · per month

Spending
$4,000 per month

WEALTH = $200,000

MONEY = $3,000
cash + checking

Everything you own, measured today

Other assets = $197,000
bonds, stocks, house, …

STOCK · measured on a date

Units test

A flow has a time unit (per month, per year). A stock is measured on a date (e.g., 31 December).

Trap 1

"She makes a lot of money." → She has a high INCOME (a flow).

Trap 2

"He has a lot of money." → He has a lot of WEALTH (a stock).

In this course

Money = cash + checking deposits: the assets you can use for transactions right away.

KEY INSIGHT

Money is a STOCK (what you hold now); income is a FLOW. Money demand = how much of your wealth you choose to hold as money.

Think-Pair-Share: What Happens to Money Demand?

ACTIVITY

1. Qatar's nominal GDP doubles due to a new LNG expansion. What happens to Md?

2. The QCB raises interest rates from 5% to 7%. What happens to Md?

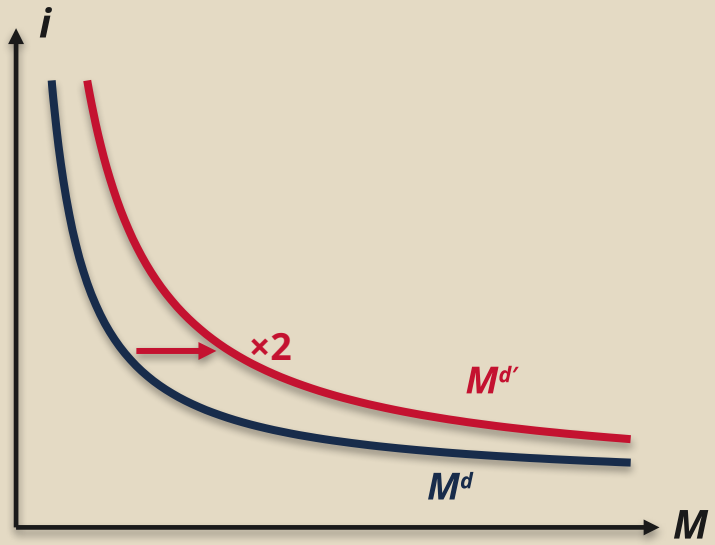
3. A new mobile payment app makes it easier to convert bonds to cash instantly. What happens to L(i)?

Discuss in pairs (3 minutes). Be ready to explain using the Md = $Y × L(i) framework.

Think-Pair-Share: Answers

Shift the curve, or move along it?

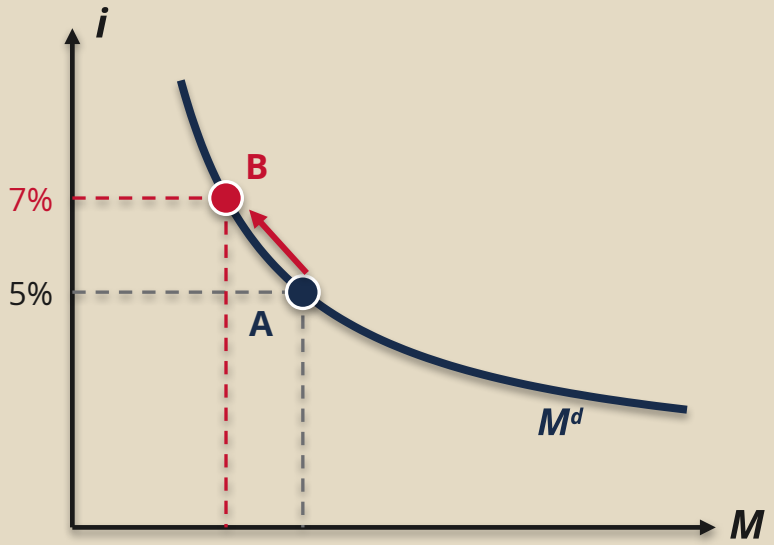
Q1 · Nominal GDP doubles (LNG expansion)



M^d shifts RIGHT

$M^d = \$Y \times L(i)$: if $\$Y$ doubles, money demand doubles at every interest rate.

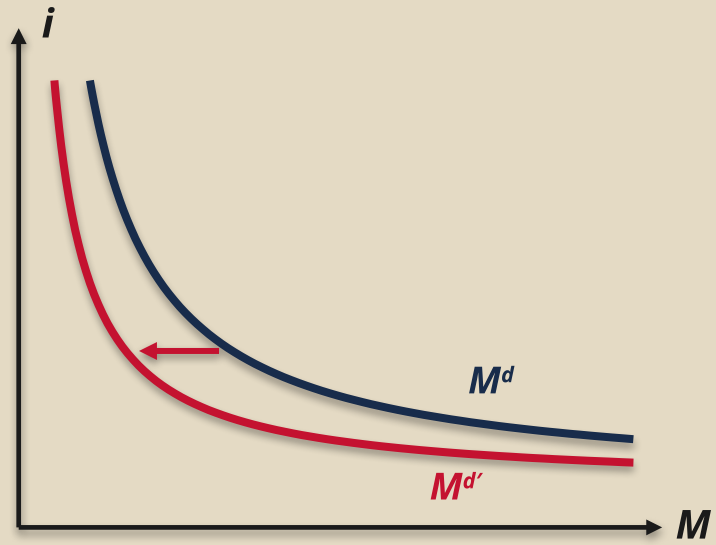
Q2 · QCB raises *i* from 5% to 7%



Movement ALONG M^d (A → B)

Higher *i* = higher opportunity cost of holding money, so people hold less. The curve itself does not move.

Q3 · Payment app: bonds → cash instantly



M^d shifts LEFT: $L(i)$ falls

If bonds turn into cash instantly, people need to hold less money at every interest rate.

$Y changes, or a new payment technology → the M^d curve SHIFTS. *i* changes → we MOVE ALONG it.

Quick Check: Money Demand

Test your understanding

True or False: Money demand rises with income.

?

True or False: Higher interest rates increase money demand.

?

If $\$Y = \$1{,}000$ and $L(i) = 0.5$, what is M^d?

?

Commit to an answer for each one before we reveal.

Quick Check: Money Demand

Test your understanding

True or False: Money demand rises with income.	**TRUE**	More income → more transactions → more money needed. M^d is proportional to $\$Y$.
True or False: Higher interest rates increase money demand.	**FALSE**	Bonds pay i, money pays 0. Higher i = higher opportunity cost, so people hold less money: $L(i)$ falls as i rises.
If $\$Y$ = \$1,000 and $L(i)$ = 0.5, what is M^d?	**\$500**	M^d = \$1,000 × 0.5 = \$500. $L(i)$ = 0.5 means people hold money worth half of their nominal income.

Follow-up 1: i rises so that $L(i) = 0.4$ → M^d = \$1,000 × 0.4 = \$400 (a movement along M^d)

Follow-up 2: $\$Y$ doubles to \$2,000 → M^d = \$2,000 × 0.5 = \$1,000 (the M^d curve shifts right)

KEY INSIGHT Money demand: proportional to \$Y, decreasing in i. The formula Md = \$Y × L(i) captures both relationships.

From Demand → Equilibrium

We know what determines money demand.
Now: How does the interest rate get determined?

Answer: Where money supply meets money demand.

SECTION 4-2

Determining the Interest Rate I

The simplest model: money supply, money demand, and equilibrium

- Money market equilibrium
- Effects of income and money supply changes
- Open market operations

Equilibrium in the Money Market

Where supply meets demand

$$Ms = Md \quad \rightarrow \quad Ms = \$Y \times L(i)$$

Money Supply (set by central bank) = Money Demand

Money Supply (Ms)

Set by the central bank as a policy decision. Vertical line on graph.

Money Demand (Md)

Determined by income ($Y) and interest rate (i). Downward-sloping curve.

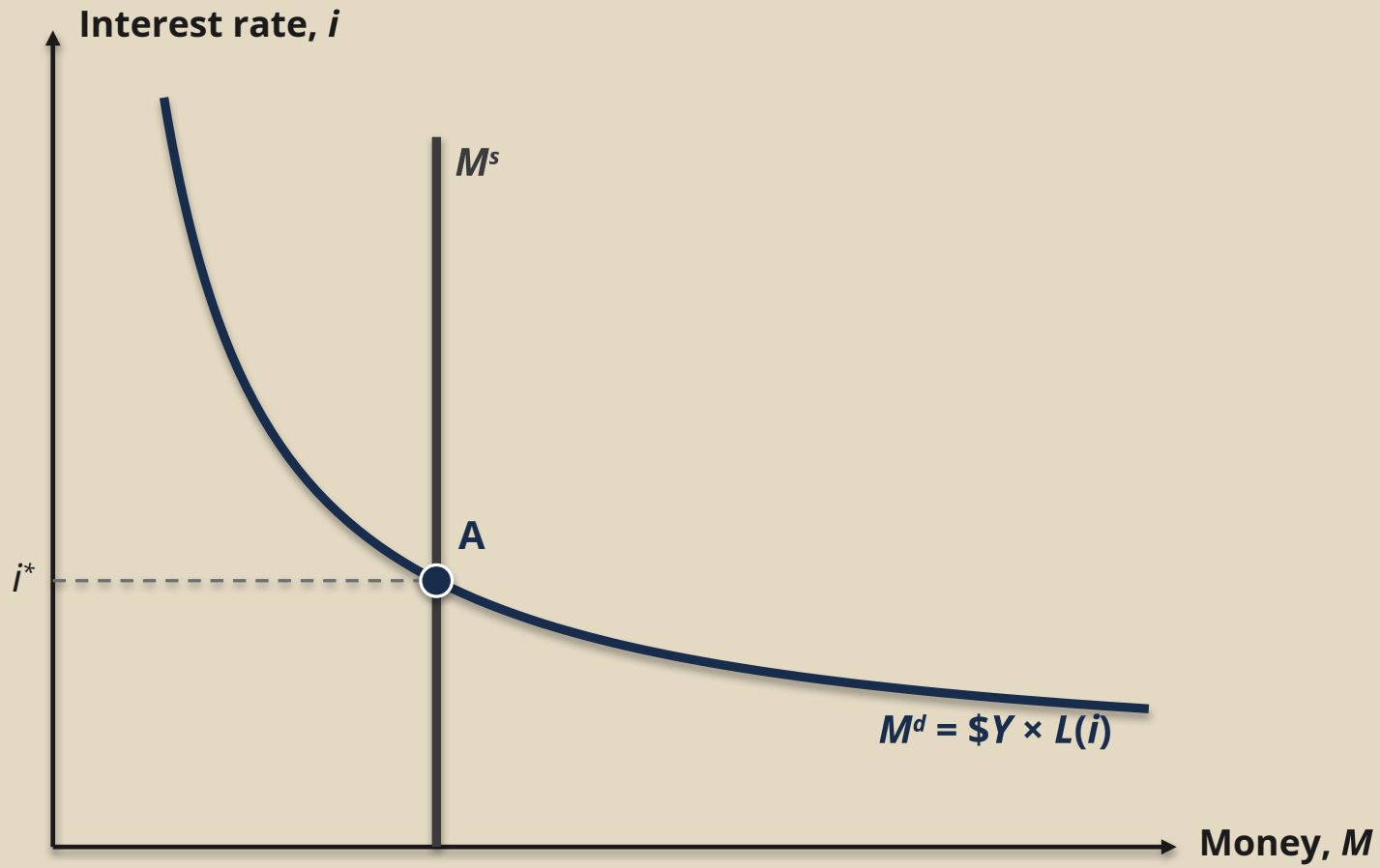
Equilibrium (i*)

The interest rate where Ms = Md. Found at intersection.

KEY INSIGHT The central bank controls Ms. Given $Y, the equilibrium interest rate i* adjusts so people willingly hold exactly Ms.

Figure 4-2: Determination of the Interest Rate

Money market equilibrium



M^d slopes down

Higher i = higher opportunity cost, so people hold less money.

M^s is vertical

The central bank fixes the money supply (a policy decision). It does not depend on i.

Equilibrium A: $M^s = M^d$

The intersection pins down the interest rate i^*.

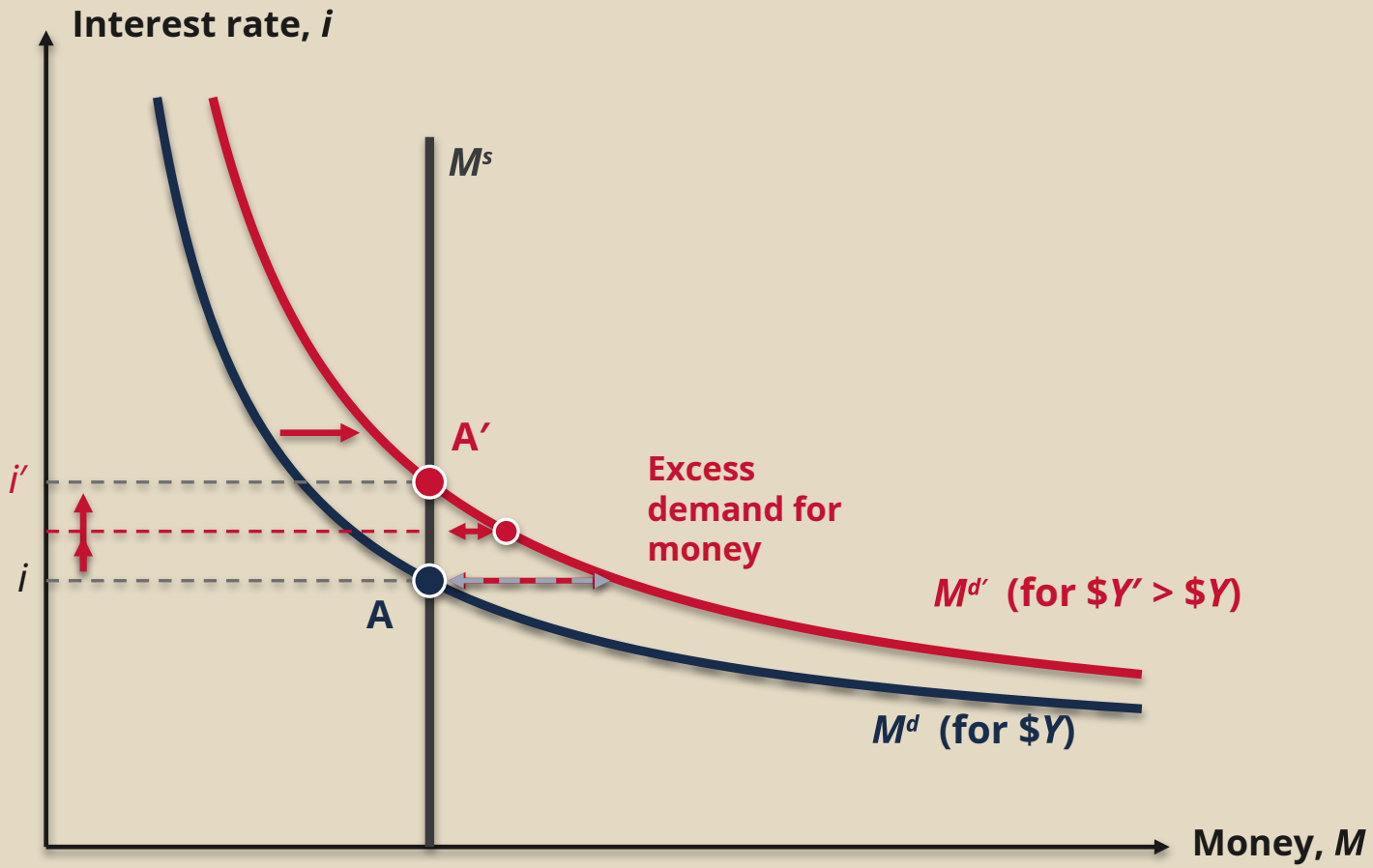
The central bank controls M^s. Given $\$Y$, i^* adjusts so people willingly hold exactly M^s.

Figure 4-3: What Happens When Income Rises?

Effect of an increase in nominal income on the interest rate

Figure 4-3: What Happens When Income Rises?

Effect of an increase in nominal income on the interest rate



Shock: income rises ($*Y* ↑)

More transactions → more money demanded at every *i*.

Adjustment through the bond market

Excess demand for money → SELL bonds

$$i = \frac{\$100 - P_B}{P_B}$$

P_B ↓

⇒ *i* ↑

P_B = price of the bond today
\$100 = what the bond pays in one year
i = the bond's yield = our interest rate

e.g. P_B falls \$96 → \$95, so *i* rises 4.2% → 5.3%

Higher *i* → holding money costs more → quantity demanded falls along $M^{d'}$.

Selling stops at A′: $M^{d'} = M^s$ at the higher i'. People willingly hold the existing money.

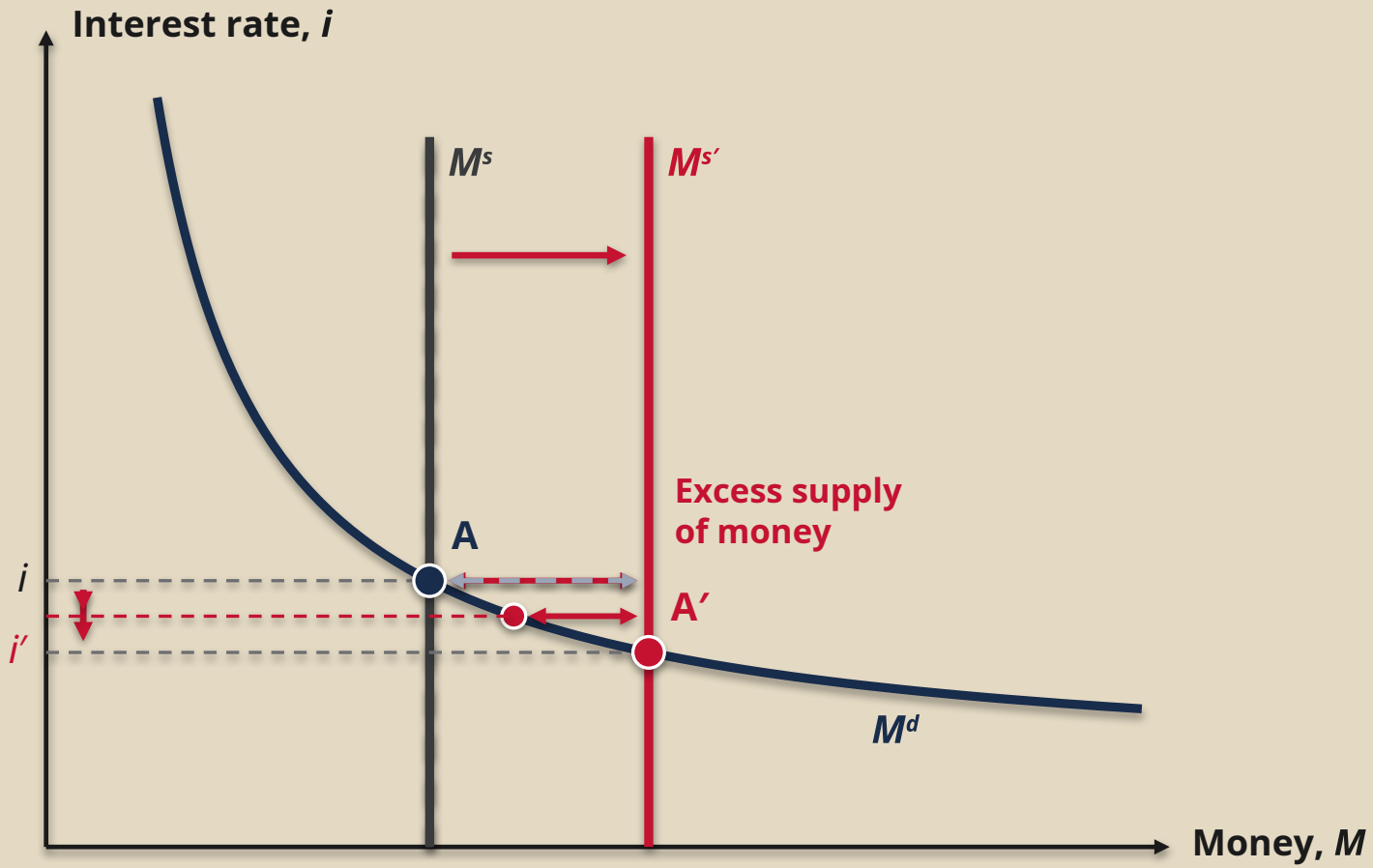
Real world: an economic boom pushes interest rates up, unless the central bank expands M^s.

Figure 4-4: What Happens When Money Supply Rises?

Effect of an increase in money supply on the interest rate

Figure 4-4: What Happens When Money Supply Rises?

Effect of an increase in money supply on the interest rate



Policy: central bank raises M^s

Money supply shifts right; money demand is unchanged.

Adjustment through the bond market

Excess supply of money → BUY bonds

$$i = \frac{\$100 - P_B}{P_B}$$

P_B ↑ ⇒ i ↓

P_B = price of the bond today
\$100 = what the bond pays in one year
i = the bond's yield = our interest rate

e.g. P_B rises \$95 → \$96, so i falls 5.3% → 4.2%

Lower i → holding money costs less → quantity demanded rises along M^d.

Buying stops at A′: $M^d = M^{s'}$ at the lower i'. People willingly hold the new money.

This is how central banks stimulate the economy!

Open Market Operations

How central banks change the money supply

The central bank's balance sheet

Assets	Liabilities
Bonds	Currency (money)
+ $1 million bonds bought	**+ $1 million** new currency

① The central bank buys $1 million of bonds from the public.

② It pays with money it creates: M^s rises by $1 million.

The public now holds more money and fewer bonds

↓

Demand for bonds ↑ → bond prices ↑

↓

Interest rate i ↓
why? next slide: $P_B = \$100 / (1 + i)$

Expansionary OMO
CB buys bonds → M^s ↑ → bond prices ↑ → i ↓

Contractionary OMO (mirror image)
CB sells bonds → M^s ↓ → bond prices ↓ → i ↑

Open market operations are how the central bank shifts M^s (Figure 4-4): buy bonds → M^s ↑ → i ↓.

Bond Prices and Interest Rates

The inverse relationship

$$P_B = \frac{\$100}{1 + i}$$

A one-year bond (T-bill) that pays $100 in one year

P_B price of the bond today

$100 face value, paid in one year

i the one-year interest rate

Same equation as in Figures 4-3 and 4-4

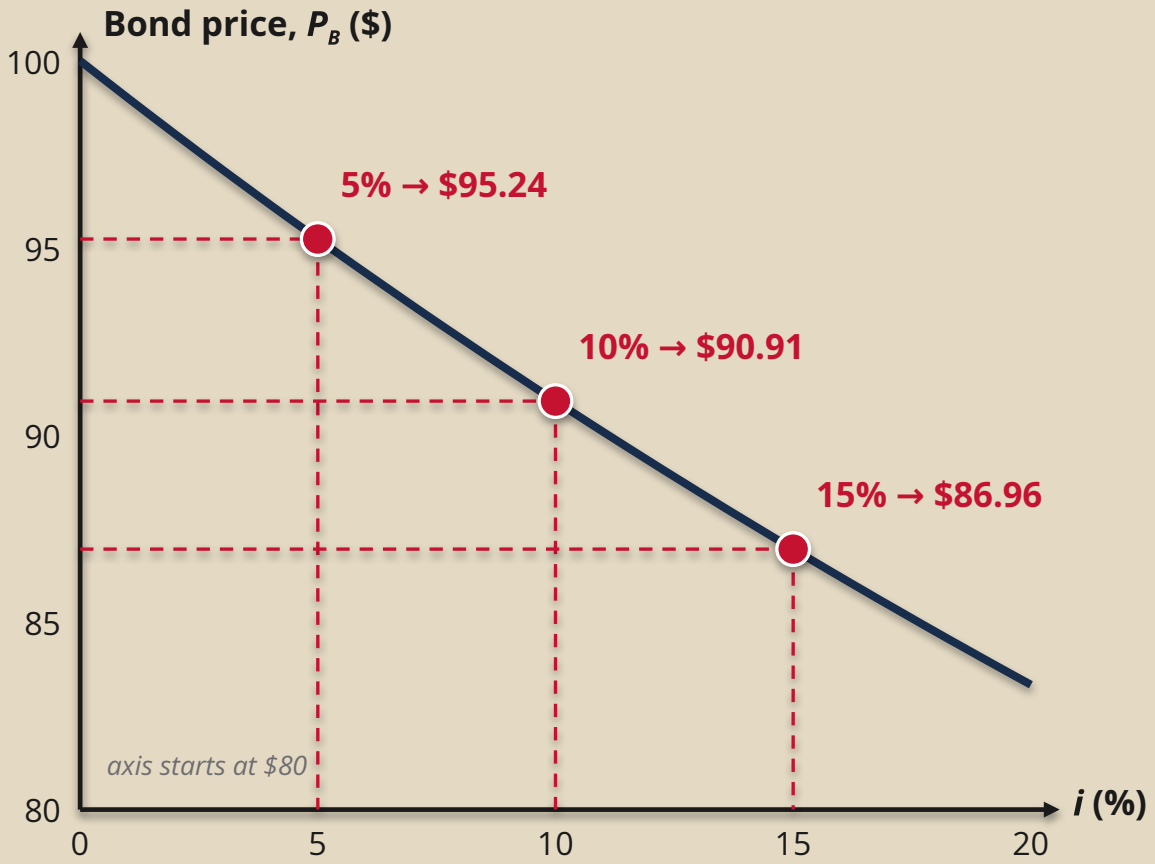
Pay P_B today, get $100 in a year.

Your interest rate is:

$$i = (\$100 - P_B) / P_B.$$

Solve for the price:

$$P_B = \$100 / (1 + i).$$



Bond price ↑ ⇔ interest rate ↓. When the central bank buys bonds, P_B rises, so i falls.

Focus: Who Holds US Currency?

Puzzle: $2.39 trillion of US cash exists. Who is holding it?

US currency in circulation

$2.39 trillion 31 Dec 2025

÷ 341.8 million people (July 2025)

≈ $7,000 in cash per person

Do you hold $7,000 in cash? Does a family of four hold $28,000? So who holds it?

Clue: 83% of that value is in $100 notes.

Sources: Federal Reserve Board, Currency in Circulation (31 Dec 2025); US Census Bureau, Vintage 2025 estimates; Federal Reserve, "The International Role of the U.S. Dollar – 2025 Edition" (FEDS Notes, Jul 2025); Fed IFDP 1387 (2024).

Where is US cash held? (Fed staff estimate, early 2025)



Estimates are uncertain: Fed researchers report a range of about 40–75%.

Why abroad?

Dollars are a safe store of value where local currencies are unstable.

Link to M^d

Foreign holdings grew more slowly as US rates rose: a higher *i* raises the opportunity cost of holding cash (Fed, 2025).

Who gains?

Cash pays no interest, so foreigners holding dollars give the US an interest-free loan (seigniorage).

KEY INSIGHT The global demand for US dollars gives the Fed extraordinary influence over world financial markets.

Case Study: Central Bank Rate Decisions

CASE STUDY

Fed rate hikes, 2022–23

To fight high inflation, the Fed raised its target range for the federal funds rate 11 times: from 0.25–0.50% (March 2022) to 5.25–5.50% (July 2023).

Q: Using our model, what did the Fed do to M^s? Draw the graph.

Discuss in groups (5 minutes). Use the M^s / M^d framework.

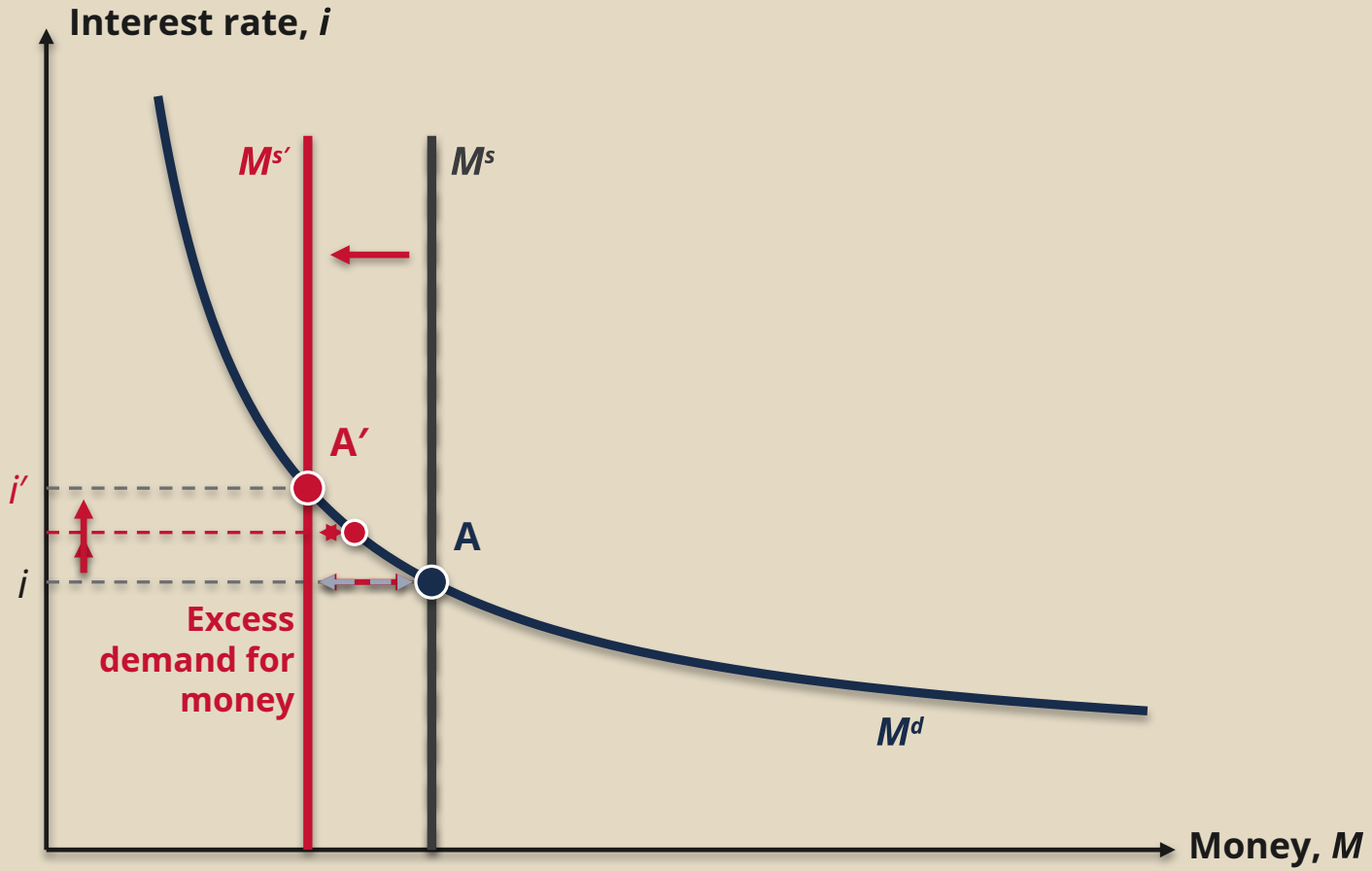
Fed funds target range, upper bound (%)



Source: Federal Reserve Board, FOMC target range history.

Case Study Solution: The Fed Tightens

Contractionary monetary policy in the money market



Example: 1-year Treasury yield, 3 Jan 2022 and 27 Jul 2023 (FRED, DGS1); price implied by $P_B = \$100/(1 + i)$.

Policy: the Fed lowers M^s

It sells bonds (contractionary OMO); money demand is unchanged.

Adjustment through the bond market

Excess demand for money → SELL bonds

$$i = \frac{\$100 - P_B}{P_B} \qquad P_B \downarrow \Rightarrow i \uparrow$$

P_B = price of the bond today
\$100 = what the bond pays in one year
i = the bond's yield = our interest rate

T-bill: P_B \$99.60 → \$94.88, so i 0.4% → 5.4%

Higher i → holding money costs more → quantity demanded falls along M^d.

Selling stops at A′: $M^d = M^{s'}$ at the higher i'. People hold the smaller money stock.

In practice the Fed announces a target range and adjusts its tools to hit it (Section 4-3). In our model, that is a leftward shift of M^s.

What Is a Bond Yield?

The return you earn on a bond

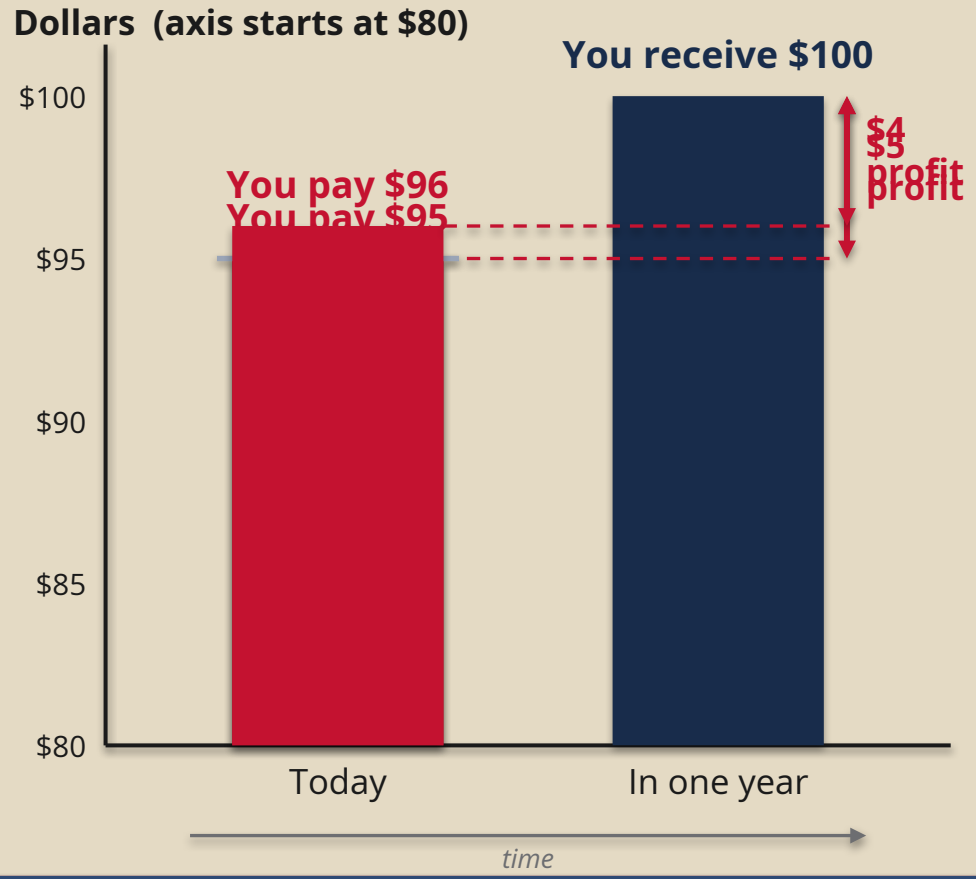
Yield = the return you earn on a bond
Plain English: your profit ÷ what you paid.

From the bond-price equation (slide 22) to the yield

1. Start: $P_B = \$100 / (1 + i)$
2. Rearrange: $1 + i = \$100 / P_B$
3. Solve: $i = (\$100 - P_B) / P_B$
4. **Pay $95: $i = \$5 / \$95 = 0.053 = 5.3\%$**

Same equation as on slides 18 and 20 (the money-market adjustment).

Think about it: if new bonds pay 7%, would anyone still pay $95 for a bond that yields 5.3%?



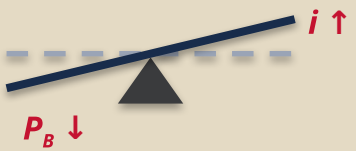
Pay $96: yield = $4 / $96 = 4.2% (price ↑ → yield ↓)

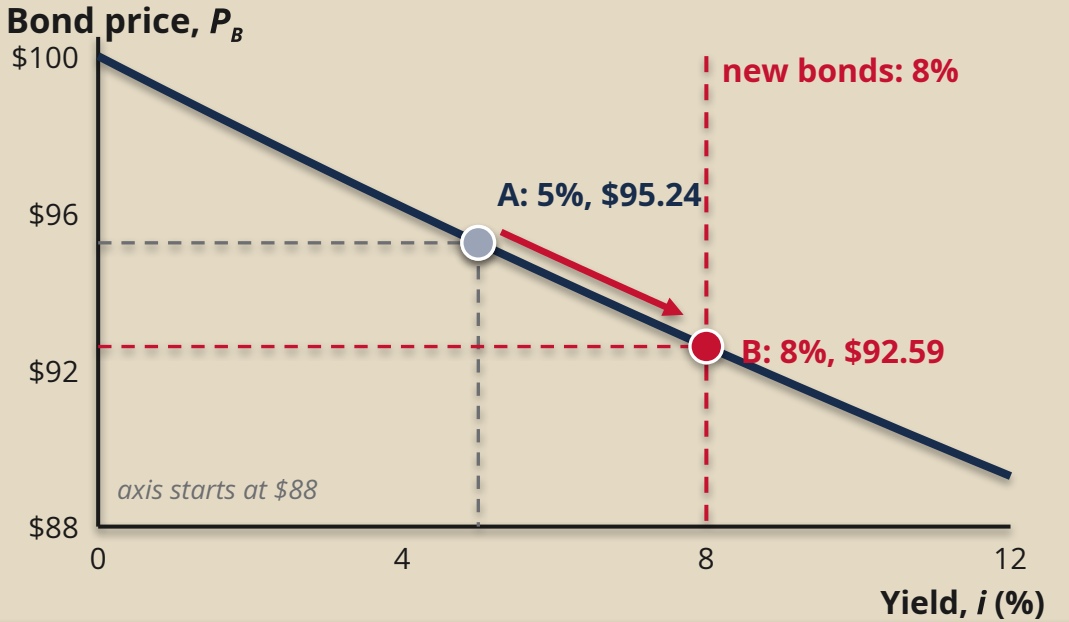
Yield = (what you get − what you pay) ÷ what you pay. Pay more for the same $100 → lower yield.

The Inverse Relationship: Bond Prices vs. Yields

Why they always move in opposite directions

1. Old bond: pays $100 in one year. Price $95.24 → yield 5%.
2. The central bank raises rates: NEW bonds now yield 8%.
3. Who pays $95.24 to earn 5% when new bonds earn 8%? Nobody. Sellers must cut the price.
4. The price falls to $92.59. Now the old bond also yields 8%: $100 / 1.08 = $92.59.
5. **The $100 never changes; only the price adjusts. Price ↓ ⇔ yield ↑.**



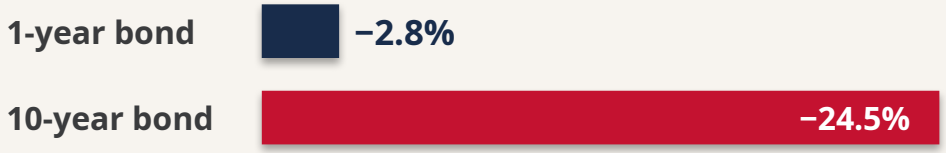


Why it matters: Silicon Valley Bank, March 2023

SVB had put depositors' money into long-term bonds bought when yields were low. As rates rose, those bonds lost value. Depositors rushed to withdraw, and the bank failed.

Same jump, 5% → 8%, for a 10-year bond

Pays $100 in 10 years: $P_B = \$100 / (1 + i)^{10}$ → $61.39 → $46.32

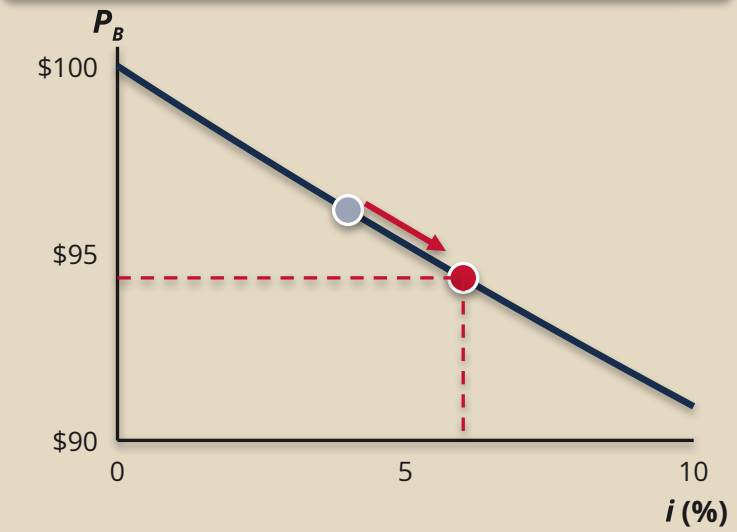


Rates up → existing bond prices down. The longer the bond, the bigger the fall.

Bond Prices & Yields in the News

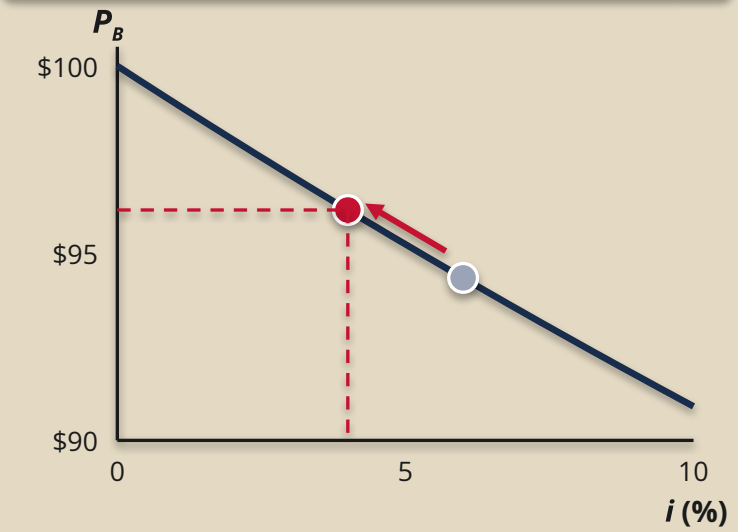
Read the headline, then move along the curve

"Treasury yields jump from 4% to 6%"



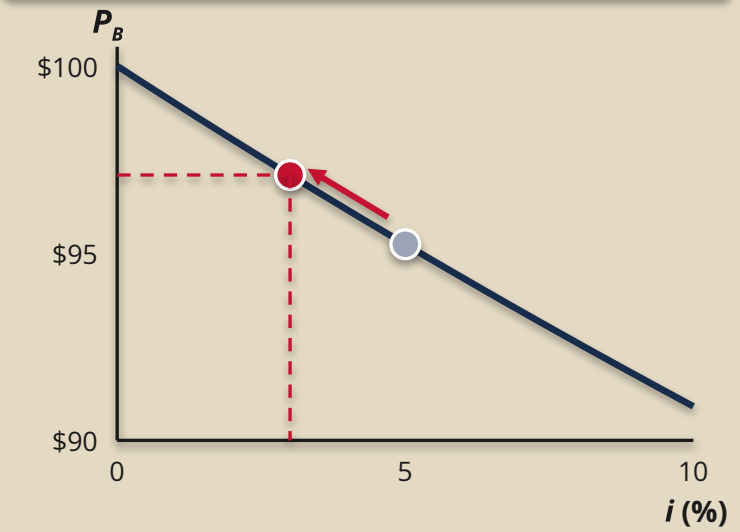
Yield ↑ ⇒ price ↓
$96.15 → $94.34. Bond prices FELL.

"Bond rally: prices climb from $94.34 to $96.15"



Price ↑ ⇒ yield ↓
6% → 4%. Bond yields FELL.

"The Fed cuts rates: new bonds now yield 3%, not 5%"



Old 5% bonds are worth more
Priced to yield 3%: $95.24 → $97.09.

Quick reference, P_B = $100 / (1 + i): 2% → $98.04 4% → $96.15 5% → $95.24 6% → $94.34 8% → $92.59 10% → $90.91

Yields and prices are two ways of reading the same bond: when one rises, the other falls.

What Is the Yield Curve?

Yields across bond maturities

The yield curve plots the yields on government bonds of different maturities, from 3 months to 30 years, on one day.

Why do longer bonds usually have HIGHER yields?

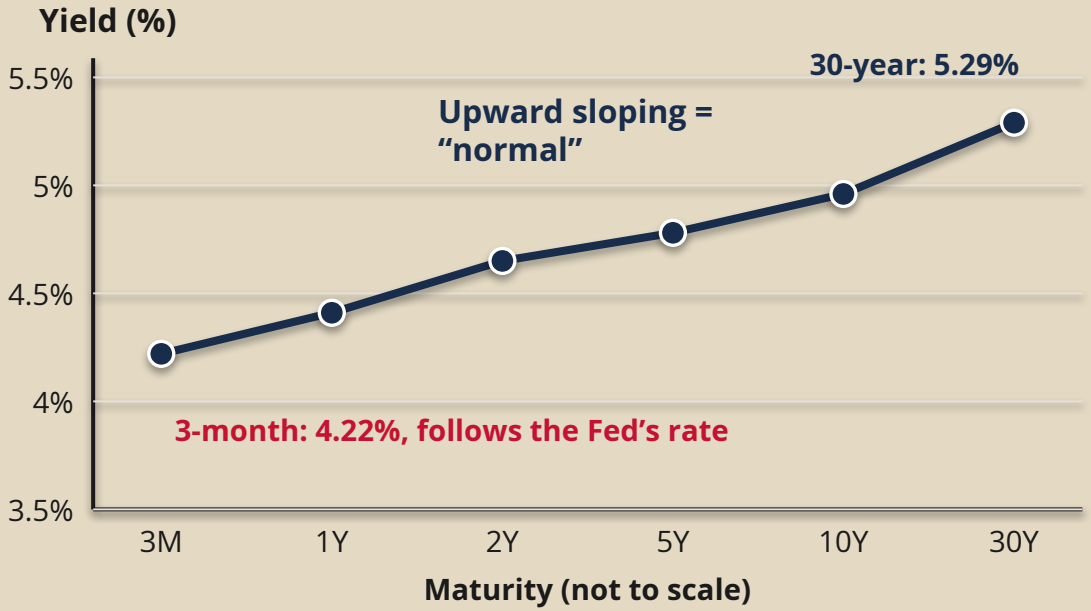
1. Expectations

A long rate ≈ the average of the short rates expected over its life. Today 4%, expected 5% next year → 2-year yield ≈ 4.5%.

2. Risk (term) premium

Long bonds are riskier to hold: their prices swing more when rates move (slide 27: −24.5% vs. −2.8%), and inflation is harder to predict far ahead. Investors want a little extra.

Usually both push the curve up. But if markets expect the Fed to CUT rates, reason 1 pulls long yields BELOW short ones → the curve inverts (next slides).



US Treasury yields, 22 Oct 2026 (FRED). Axis starts at 3.5%.

Blanchard 14-2: the 2-year yield



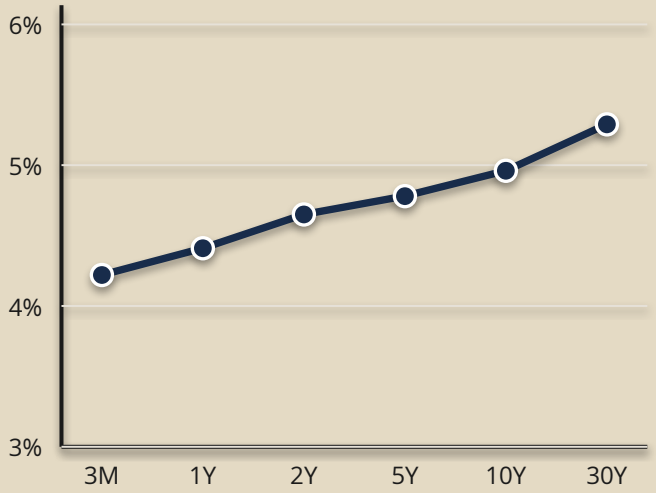
Short end: the Fed's rate today. Long end: what markets expect the Fed to do later, plus a premium.

The Three Shapes of the Yield Curve

Real US Treasury curves. What does each one signal?

Normal (upward)

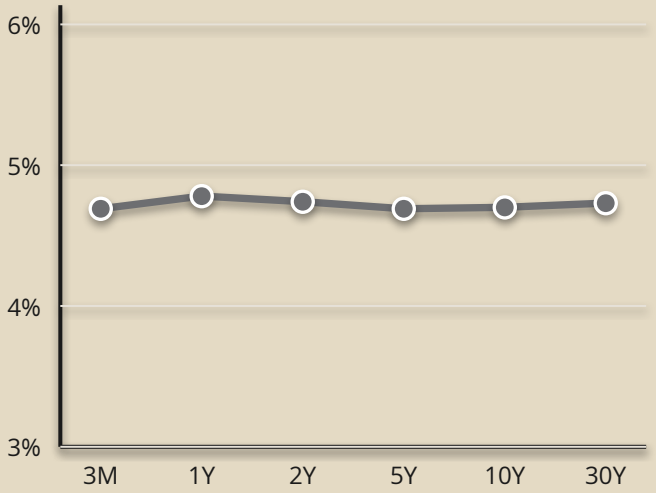
22 Oct 2026



Signal?

Flat

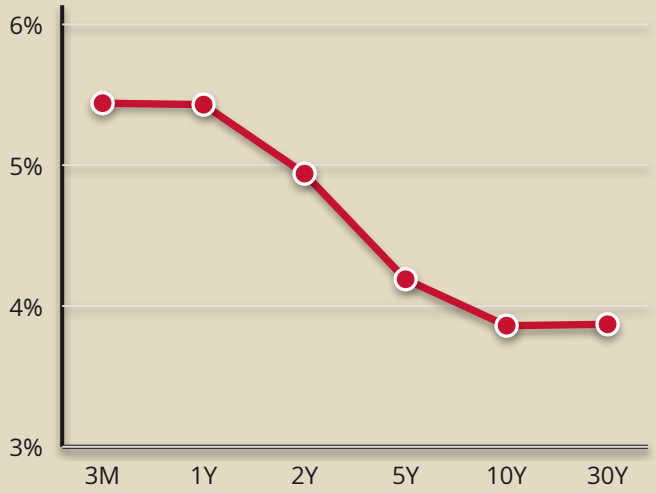
22 Mar 2006



Signal?

Inverted (downward)

3 Jul 2023



Signal?

Pairs (2 minutes): for each shape, where do markets expect the Fed's rate to go?

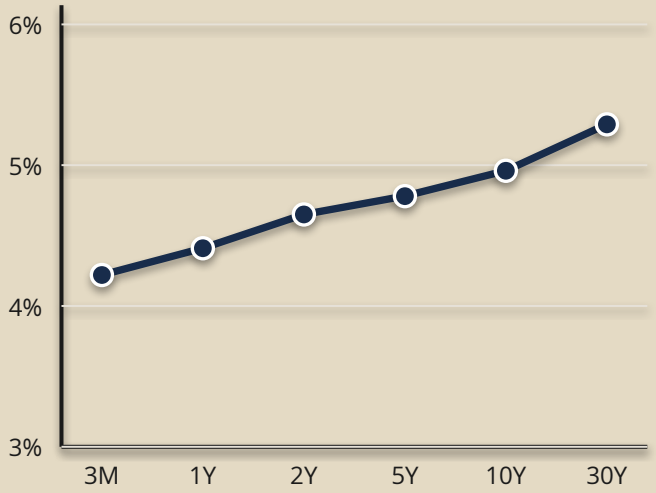
Source: FRED, US Treasury constant-maturity yields (DGS3MO … DGS30). Yield axis 3–6%, maturities not to scale.

The Three Shapes of the Yield Curve

Each shape is a forecast of the Fed's future rate

Normal (upward)

22 Oct 2026

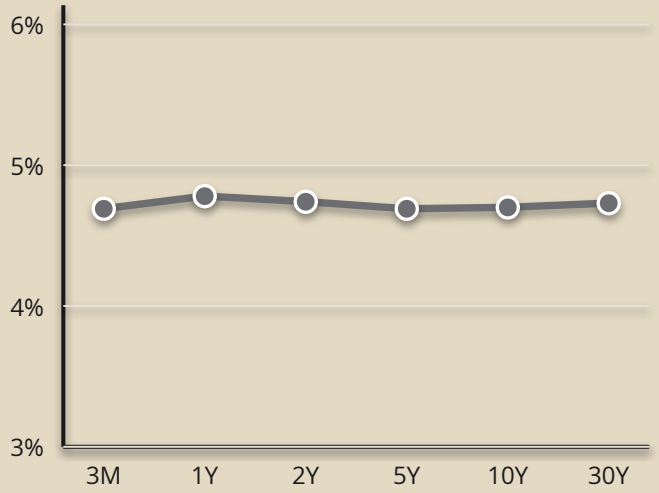


Expected Fed rate: rise or stay

Normal times
Short rates expected to rise or stay, plus a premium → long > short.

Flat

22 Mar 2006

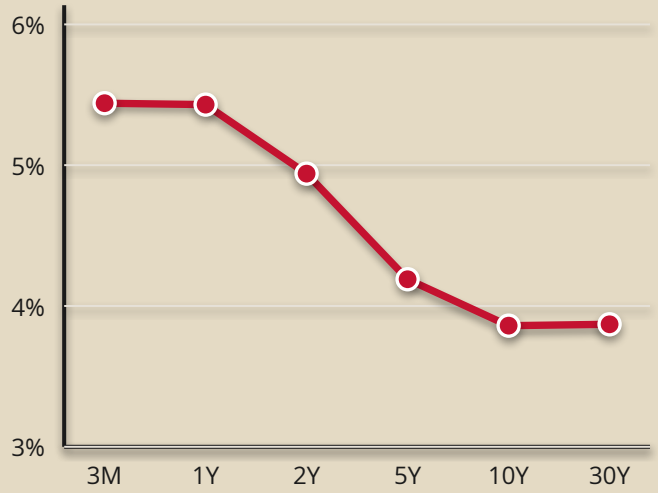


Expected Fed rate: stay put

A turning point
Short rates expected to stay put. Mar 2006: near the end of the 2004–06 hikes.

Inverted (downward)

3 Jul 2023



Expected Fed rate: fall

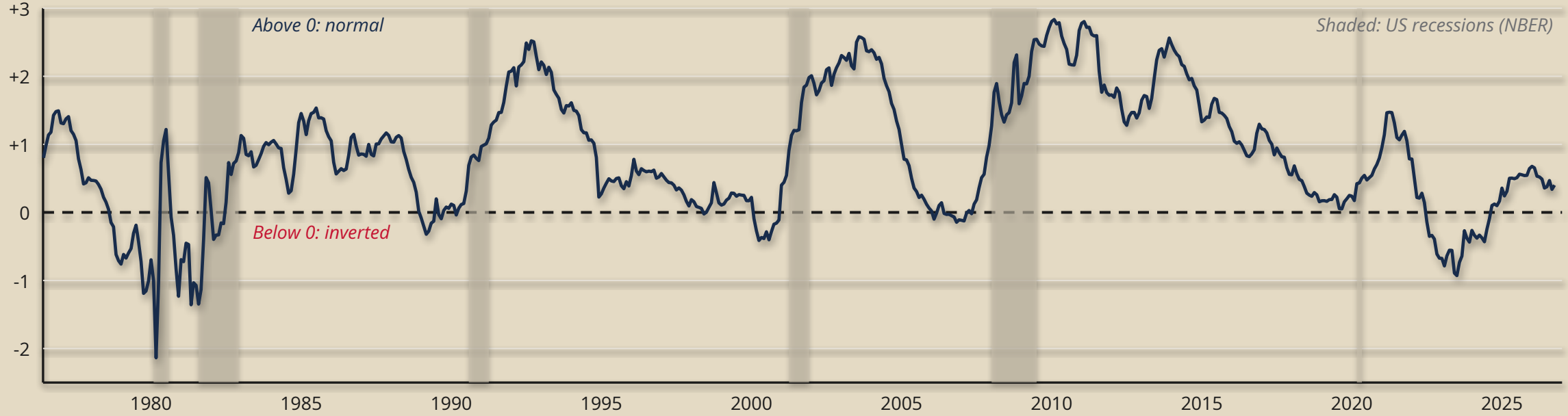
Warning sign
Markets expect Fed CUTS, often because a slowdown is coming → long < short.

Upward = short rates expected to rise or stay (plus a premium); inverted = cuts expected.

The Inverted Yield Curve: A Recession Signal?

Fifty years of US data

10-year minus 2-year Treasury yield (percentage points)



Source: FRED, T10Y2Y (monthly average of daily data, Jun 1976–Oct 2026) and USREC.

What do you see just BEFORE each shaded recession?

Look at where the line is relative to 0.